

Problem Set 8 — Munson

Replication Cooper & Ejarque (2003)

Replication Blueprint: Code Modules Needed

Organizational plan for the computational implementation.

1. Module A: Parameter Initialization

Store fixed parameters (β, δ, p) , initial guesses for structural parameters $(\alpha, \gamma, \rho, \sigma, \tilde{\phi}_0)$, and grid sizes.

2. Module B: Tauchen Discretization of the Shock Process

Construct the Markov chain for the profitability shock A using Tauchen's method: produce grid points and transition matrix P_A .

3. Module C: Capital Grid Construction

Create a grid for capital K (e.g., 600 points for the costly external finance model). Define $K_{\min}$, $K_{\max}$, and spacing.

4. Module D: Profit and Adjustment Cost Functions

Implement

$$\pi(K, A) = AK^\alpha, \quad C(K', K) = \frac{\gamma}{2} \left(\frac{K' - (1 - \delta)K}{K} \right)^2 K.$$

5. Module E: Bellman Solver

Solve the dynamic optimization problem allowing for two cases:

- internal finance (no external cost),
- external finance (pay fixed cost $\tilde{\phi}_0$).

For each state (K, A) compute value and choose the maximizing K' and regime.

6. Module F: Compute Policy-Implied Variables

From $K'(K, A)$ compute investment I , average $Q = V(K, A)/K$, profits, and external-finance indicator.

7. Module G: Panel Simulation

Simulate a panel of firms (e.g. 1000 firms $\times$ 50 periods) using the policy functions and Markov process for A .

8. Module H: Compute Simulated Moments

From the simulated panel compute:

- a_1, a_2 from the Q regression,

- serial correlation of I/K ,
- standard deviation of π/K ,
- mean average Q ,
- fraction of externally financed investment.

9. Module I: SMM Objective Function

Construct

$$J(\Theta) = (\Psi_d - \Psi_s(\Theta))' W (\Psi_d - \Psi_s(\Theta)) ,$$

using target moments Ψ_d from the paper and simulated moments $\Psi_s(\Theta)$.

10. Module J: Optimization Routine

Search over Θ to minimize $J(\Theta)$ using global+local methods.

11. Module K: Standard Errors (Optional)

Estimate covariance of simulated moments and apply SMM sandwich formula.

12. Module L: Output & Table Construction

Format the estimated parameters and simulated moments to reproduce Table 3.

Unable to converge

Unfortunately I could not get my code to converge to the papers parameters. I've tried implementing multiple tests and fixes but have been unable to find a solution to make the code converge.